PREDICTIVE ANALYTICS FOR CREDIT RISK MANAGEMENT: ENHANCING DEFAULT PREDICTION IN BANKING

BY UTILIZING SUPERVISED LEARNING TO PREDICT DEFAULT RISKS AND OPTIMIZE LENDING PRACTICES

Great Learning
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BUSINESS PROBLEM STATEMENT

1. To develop a predictive model for a bank to forecast the probability of default among its customers.
2. To explore the relationships between various customer variables and credit default to improve predictive accuracy.
3. Implementing a supervised learning classification approach to accurately categorize customers as potential defaulters or non-defaulters.
4. Evaluate the effectiveness of the predictive model in identifying default risks and suggest lending practices to banks.
5. Utilizing predictive analytics, we aim to mitigate financial losses resulting from default events by effectively managing credit exposure.

METHODOLOGY

Data Preprocessing

- Data cleaning
- Exploratory data analysis (EDA)

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Model Building

- Model training
- Model testing

↓

Model analysis and Inferences

- Model tuning
- Business case analysis & Recommendations

INFORMATION ABOUT THE DATASET

- Shape of the dataset : Rows – 99,979,Columns – 36
- Variables datatypes : Float (Numerical) – 33,
- Object (Categorical) – 3
- Total missing values in the dataset : 700141
- Below mentioned are the variables with missing values more than 49% in the provided dataset

worst_status_12_24m	66763
worst_status_6_12m	60352
debt_vs_paid_ratio_0_24m	59317
worst_status_3_6m	57704
worst_status_0_3m	54375
account_status	54375
avg_payment_span_0_3m	49307

```
RangeIndex: 99979 entries, 0 to 99978
Data columns (total 36 columns):
 #   Column                            Non-Null Count  Dtype
---  ------                            --------------  -----
 0   userid                            99977 non-null  float64
 1   default                           89977 non-null  float64
 2   acct_amt_added_12_24m             99977 non-null  float64
 3   acct_days_in_dc_12_24m            88141 non-null  float64
 4   acct_days_in_rem_12_24m           88141 non-null  float64
 5   acct_days_in_term_12_24m          88141 non-null  float64
 6   acct_incoming_debt_vs_paid_0_24m  40662 non-null  float64
 7   acct_status                       45604 non-null  float64
 8   acct_worst_status_0_3m            45604 non-null  float64
 9   acct_worst_status_12_24m          33216 non-null  float64
 10  acct_worst_status_3_6m            42275 non-null  float64
 11  acct_worst_status_6_12m           39627 non-null  float64
 12  age                               99977 non-null  float64
 13  avg_payment_span_0_12m            76141 non-null  float64
 14  avg_payment_span_0_3m             50672 non-null  float64
 15  merchant_category                 99977 non-null  object
 16  merchant_group                    99968 non-null  object
 17  has_paid                          88943 non-null  float64
 18  max_paid_inv_0_12m                88943 non-null  float64
 19  max_paid_inv_0_24m                88943 non-null  float64
 20  name_in_email                     88943 non-null  object
 21  num_active_div_by_paid_inv_0_12m  70052 non-null  float64
 22  num_active_inv                    88943 non-null  float64
 23  num_arch_dc_0_12m                 88943 non-null  float64
 24  num_arch_dc_12_24m                88943 non-null  float64
 25  num_arch_ok_0_12m                 88943 non-null  float64
 26  num_arch_ok_12_24m                88943 non-null  float64
 27  num_arch_rem_0_12m                88943 non-null  float64
 28  status_max_archived_0_6_months    88943 non-null  float64
 29  status_max_archived_0_12_months   88943 non-null  float64
 30  status_max_archived_0_24_months   88943 non-null  float64
 31  recovery_debt                     88943 non-null  float64
 32  sum_capital_paid_acct_0_12m       88943 non-null  float64
 33  sum_capital_paid_acct_12_24m      88943 non-null  float64
 34  sum_paid_inv_0_12m                88943 non-null  float64
 35  time_hours                        88943 non-null  float64
dtypes: float64(33), object(3)
```

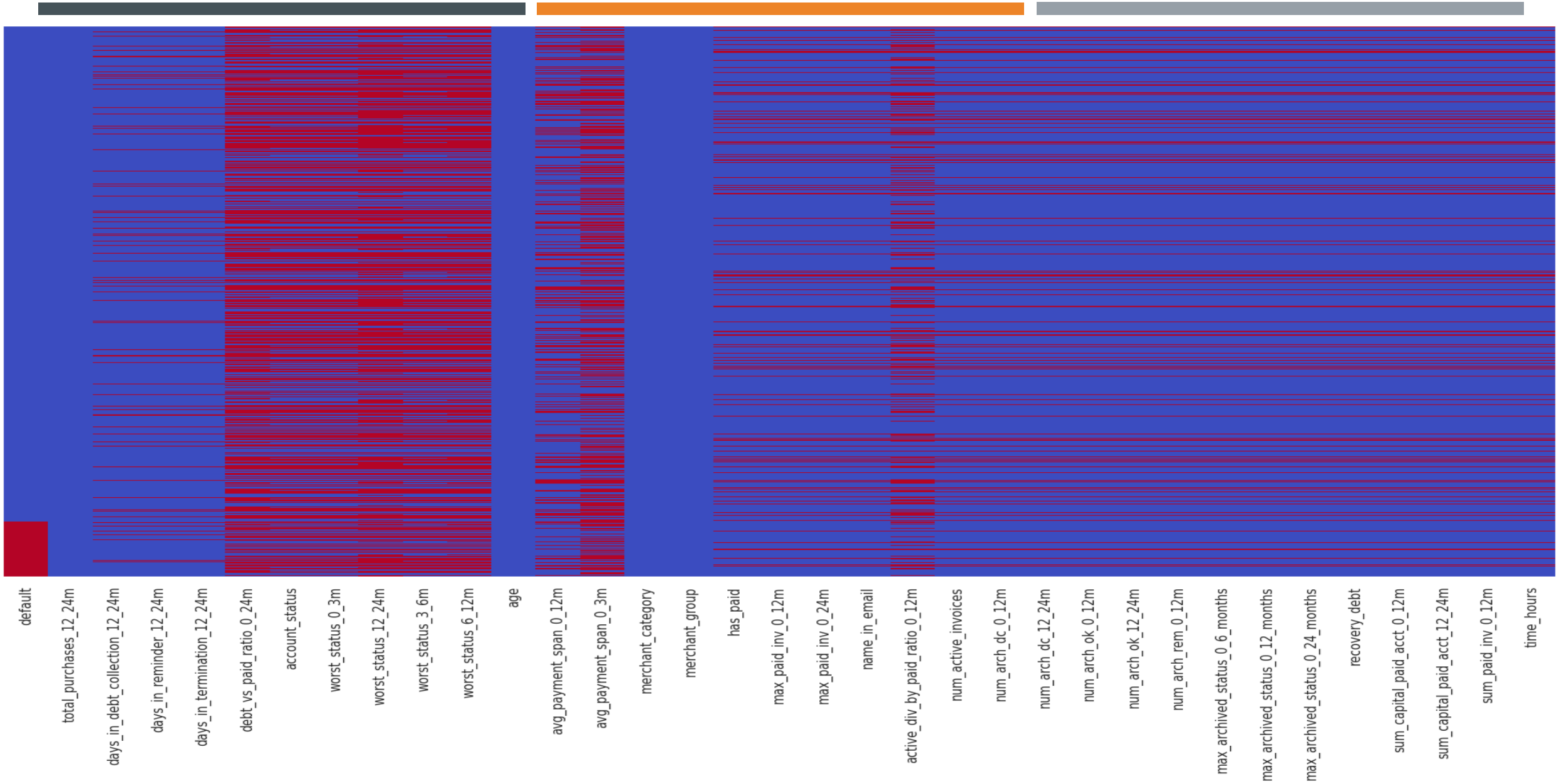

Figure 1: Missing values representation in heat map

DATA CLEANING – MISSING VALUES TREATMENT

- Approach – Removing the columns with missing value more than 49%
- (worst_status_12_24m, worst_status_6_12m, debt_vs_paid_ratio_0_24m, worst_status_3_6m worst_status_0_3m, account_status, avg_payment_span_0_3m)
- Updated dataset - (99979, 29)
- Then the null values in updated dataset was treated with mean values for numerical values and categorical values are mode values
- Output of dataset post treating the missing values is mentioned→

default	0
total_purchases_12_24m	0
days_in_debt_collection_12_24m	0
days_in_reminder_12_24m	0
days_in_termination_12_24m	0
age	0
avg_payment_span_0_12m	0
merchant_category	0
merchant_group	0
has_paid	0
max_paid_inv_0_12m	0
max_paid_inv_0_24m	0
name_in_email	0
active_div_by_paid_ratio_0_12m	0
num_active_invoices	0
num_arch_dc_0_12m	0
num_arch_dc_12_24m	0
num_arch_ok_0_12m	0
num_arch_ok_12_24m	0
num_arch_rem_0_12m	0
max_archived_status_0_6_months	0
max_archived_status_0_12_months	0
max_archived_status_0_24_months	0
recovery_debt	0
sum_capital_paid_acct_0_12m	0
sum_capital_paid_acct_12_24m	0
sum_paid_inv_0_12m	0
time_hours	0
log_sum_paid_inv_0_12m	0

DATA CLEANING – TREATING OUTLIERS

- Approach involves detection of outliers in each numerical variables in the dataset using Z-score (setting the threshold value as 3) based outlier detection technique.
- Winsorization technique is used to treat the outliers
- Total outliers detected in each variable is mentioned→

```
Number of outliers in each column:
default                              1289
total_purchases_12_24m               1228
days_in_debt_collection_12_24m         52
days_in_reminder_12_24m               662
days_in_termination_12_24m              1
age                                   433
avg_payment_span_0_12m                  2
has_paid                                0
max_paid_inv_0_12m                   1687
max_paid_inv_0_24m                   1687
active_div_by_paid_ratio_0_12m          0
num_active_invoices                  1287
num_arch_dc_0_12m                     960
num_arch_dc_12_24m                    961
num_arch_ok_0_12m                    2051
num_arch_ok_12_24m                   1809
num_arch_rem_0_12m                   1508
max_archived_status_0_6_months       2037
max_archived_status_0_12_months      2037
max_archived_status_0_24_months      2037
recovery_debt                         192
sum_capital_paid_acct_0_12m          2151
sum_capital_paid_acct_12_24m         2166
sum_paid_inv_0_12m                   1228
time_hours                           1067
log_sum_paid_inv_0_12m                  0
```

```
Updated count of outliers after treatment:
 active_div_by_paid_ratio_0_12m        0
age                                    0
avg_payment_span_0_12m                 0
days_in_debt_collection_12_24m         0
days_in_reminder_12_24m                0
days_in_termination_12_24m             0
default                             1289
has_paid                               0
log_sum_paid_inv_0_12m                 0
max_archived_status_0_12_months        0
max_archived_status_0_24_months        0
max_archived_status_0_6_months         0
max_paid_inv_0_12m                     0
max_paid_inv_0_24m                     0
merchant_category                      0
merchant_group                         0
name_in_email                          0
num_active_invoices                    0
num_arch_dc_0_12m                      0
num_arch_dc_12_24m                     0
num_arch_ok_0_12m                      0
num_arch_ok_12_24m                     0
num_arch_rem_0_12m                     0
recovery_debt                          0
sum_capital_paid_acct_0_12m            0
sum_capital_paid_acct_12_24m           0
sum_paid_inv_0_12m                     0
time_hours                             0
total_purchases_12_24m                 0
```

DATA CLEANING – TREATING OUTLIERS

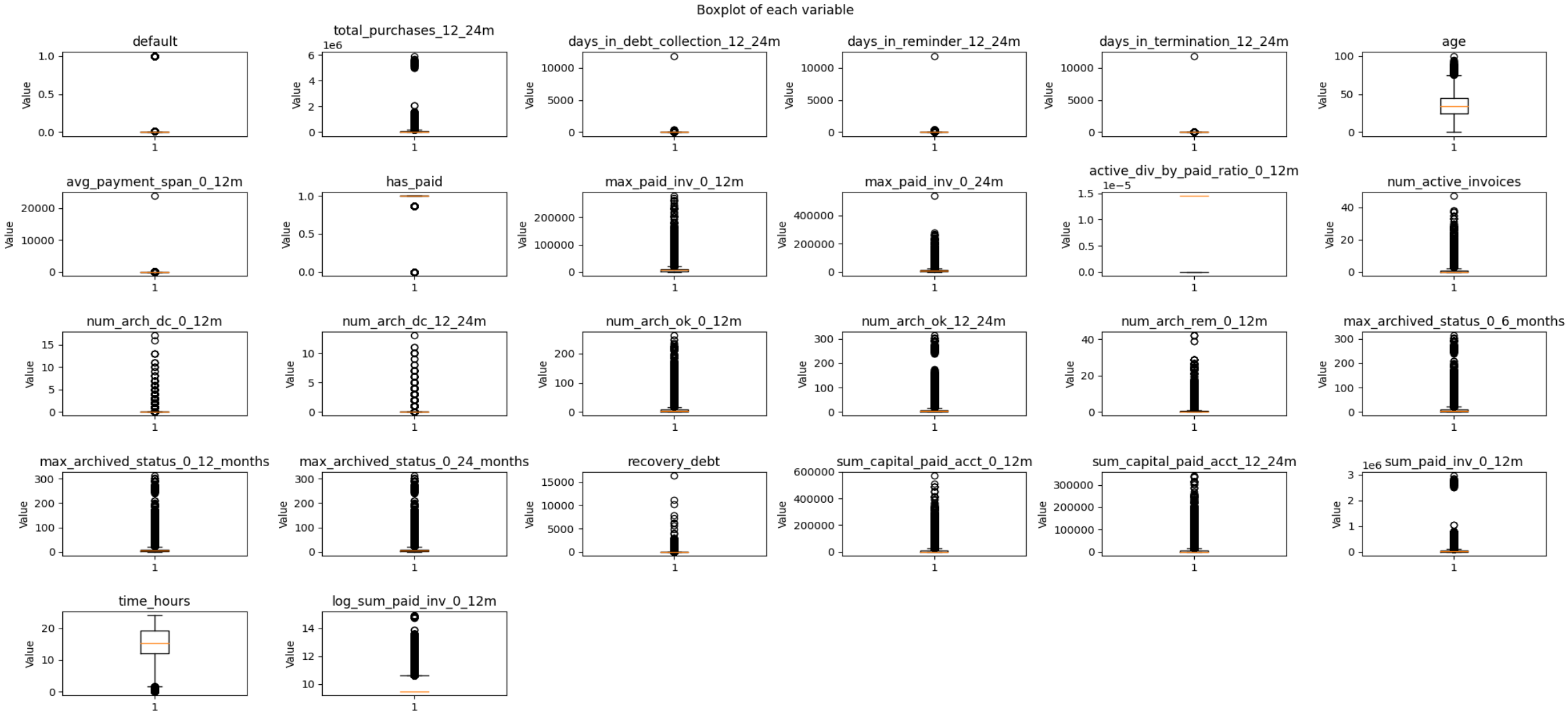

Figure 2: Outlier representation in boxplots for numerical variables

EXPLORATORY DATA ANALYSIS – DISTRIBUTION PLOTS

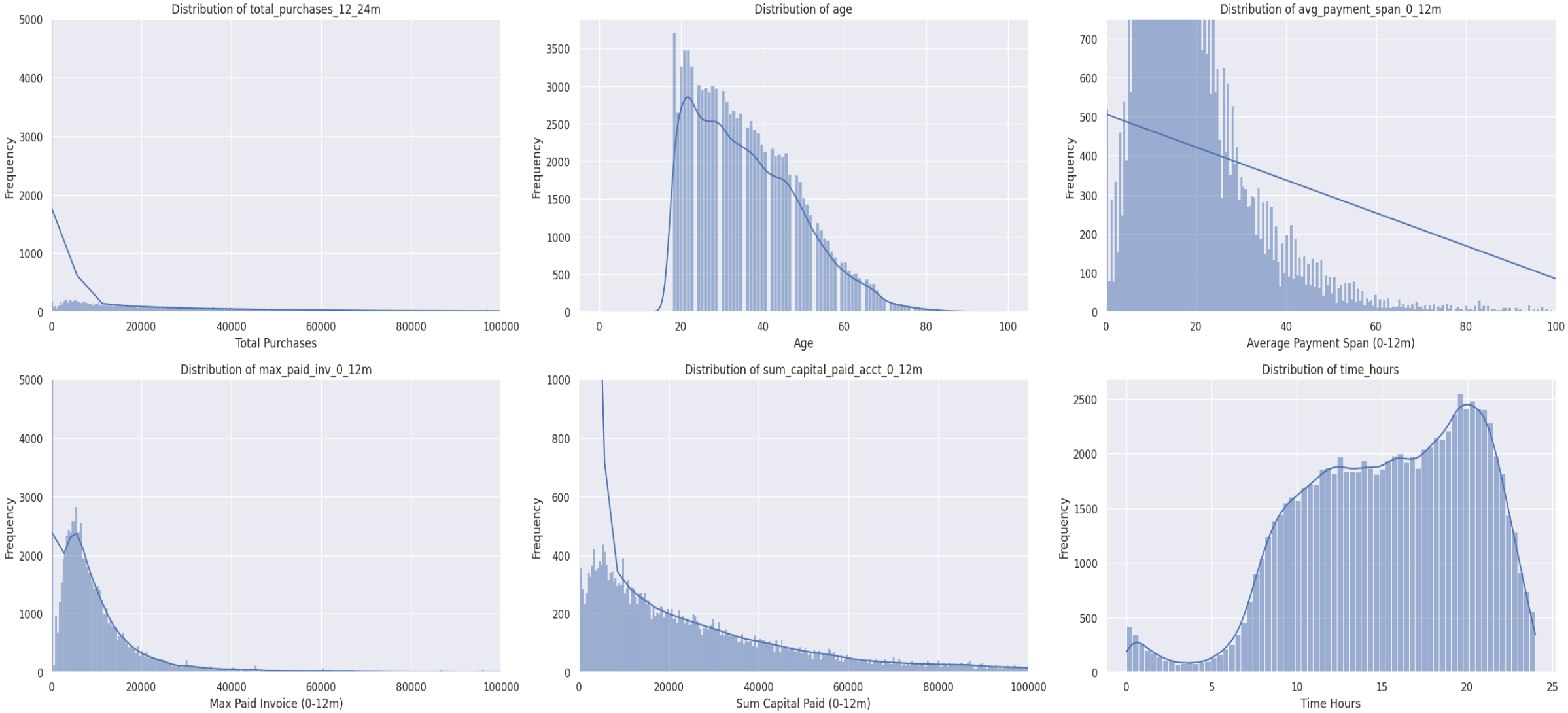

Figure 3: Distribution histogram plots – Total Purchases, Age, Avg payment span (0-12m), Max paid invoice (0-12m), sum capital paid ,Time hours

EXPLORATORY DATA ANALYSIS - PAIR PLOT

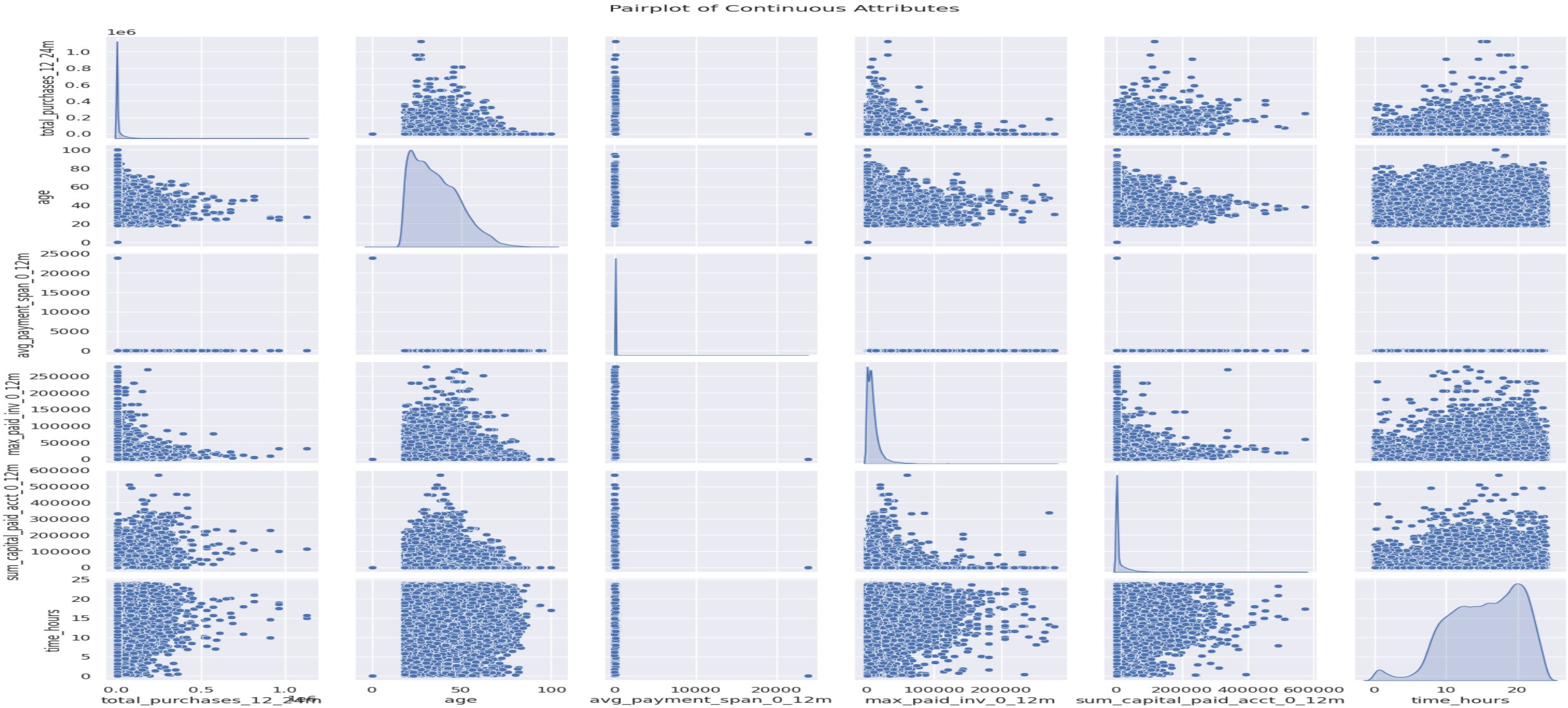

Figure 4: Pair plots – Total Purchases, Age, Avg payment span (0-12m), Max paid invoice (0-12m), sum capital paid ,Time hours

EXPLORATORY DATA ANALYSIS – CORRELATION HEATMAP

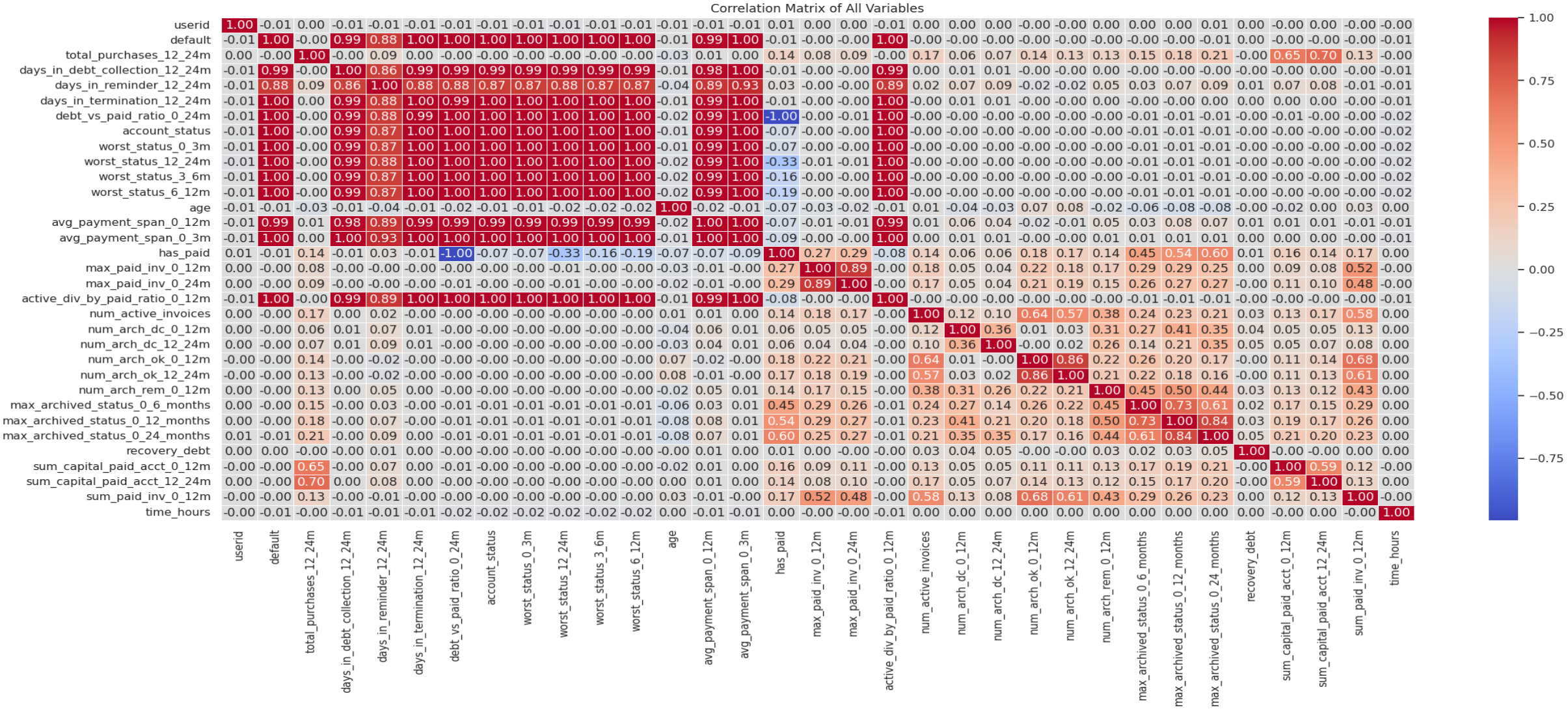

Figure 5: Correlation heatmap for variables

EXPLORATORY DATA ANALYSIS – CATEGORICAL VARAIBLES

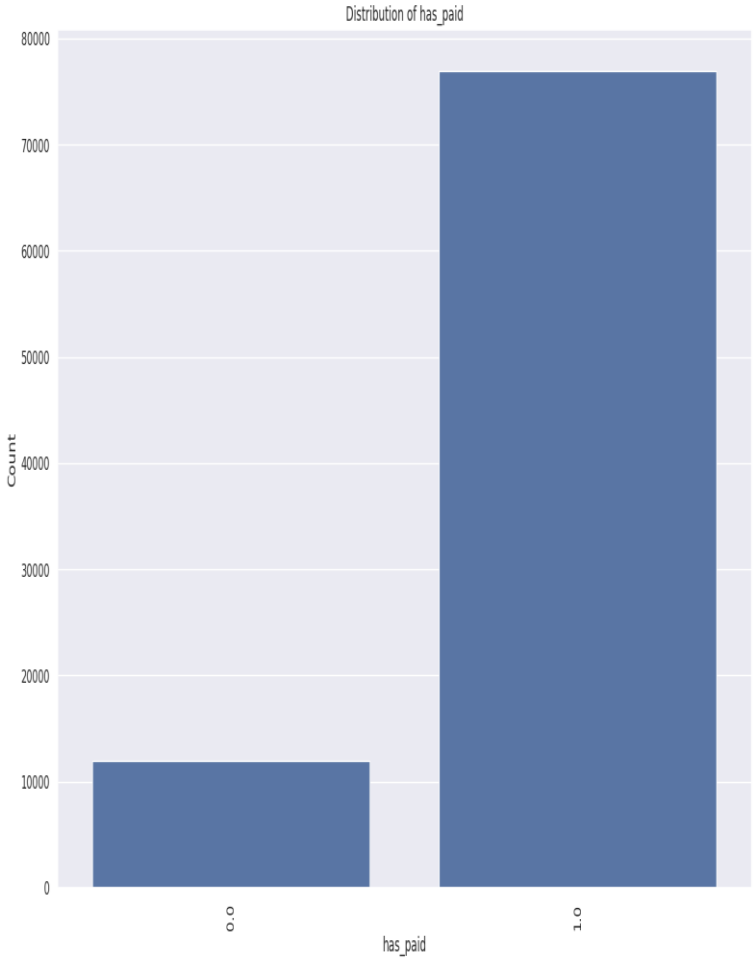

Figure 6: Histogram for has paid

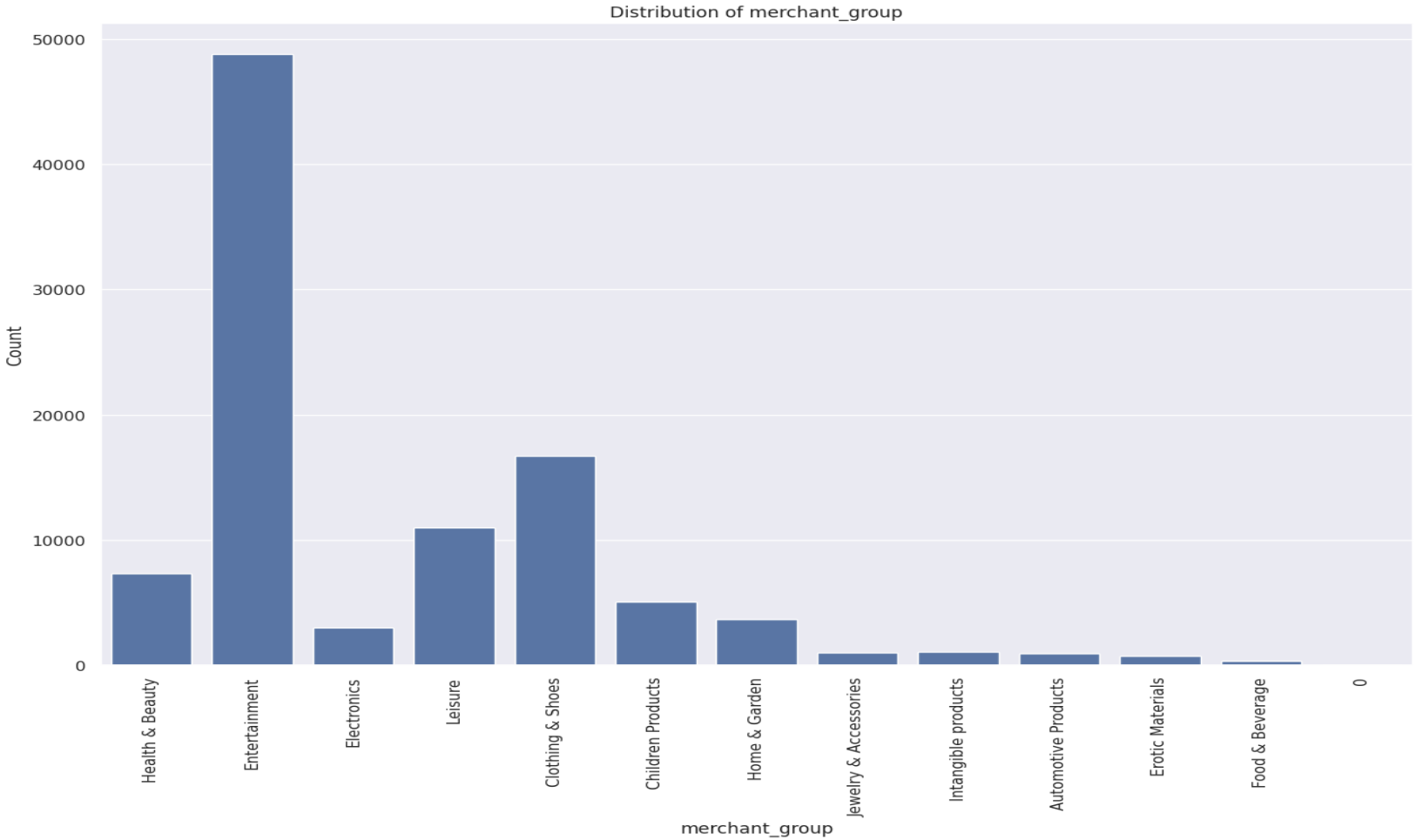

Figure 7: Histogram for merchant group

EXPLORATORY DATA ANALYSIS – CATEGORICAL VARAIBLES

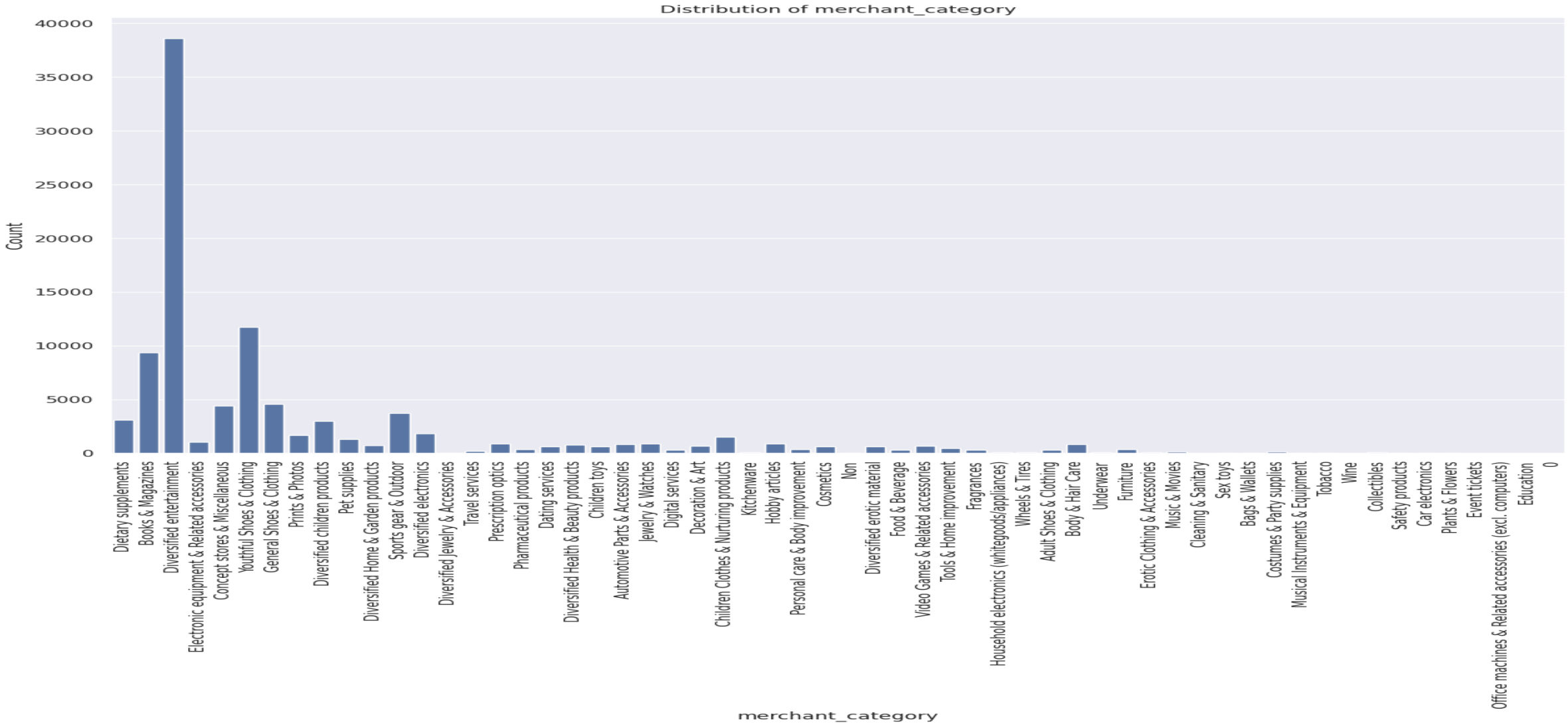

Figure 8: Histogram for merchant category

PREPARING DATASET FOR MODEL TRAINING

- Post treating the outliers some of the rows is replaced with (inf and –inf) which is dropped from the treated dataset for model training and the updated shape of the dataset is (71096, 29)
- Default is selected as the target variable
- Numerical feature data is standardized before training the model
- Categorical data is transformed into numerical data using the 'OneHotEncoder' technique for modeling
- Now the data is split in train and test data (70:30)

Training data shape (X_train, y_train): (49767, 92) (49767,)
Testing data shape (X_test, y_test): (21329, 92) (21329,)

Name of the model

1. Logistic Regression model
2. Decision Tree Classifier model
3. Random Forest Classifier model
4. AdaBoost Classifier model
5. Gradient Boosting Classifier model
6. Bagging Classifier model
7. Light GBM Classifier model
8. XGB Classifier model
9. Linear Discriminant Analysis model

TEST SUMMARY - ROC CURVE – ALL MODELS

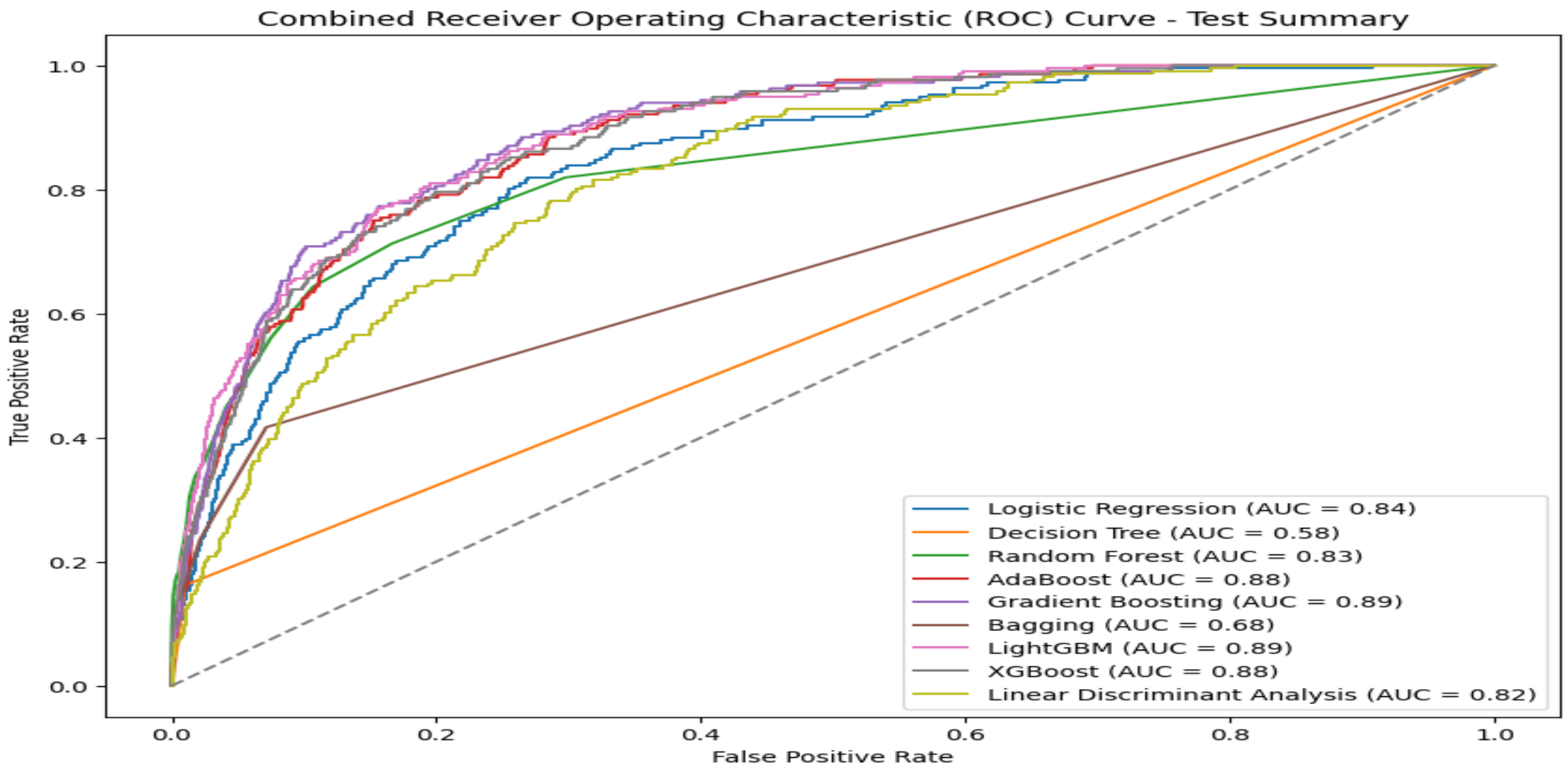

Figure 9: Combined ROC curve for test summary – All models

TEST SUMMARY – PRECISION-RECALL – ALL MODELS

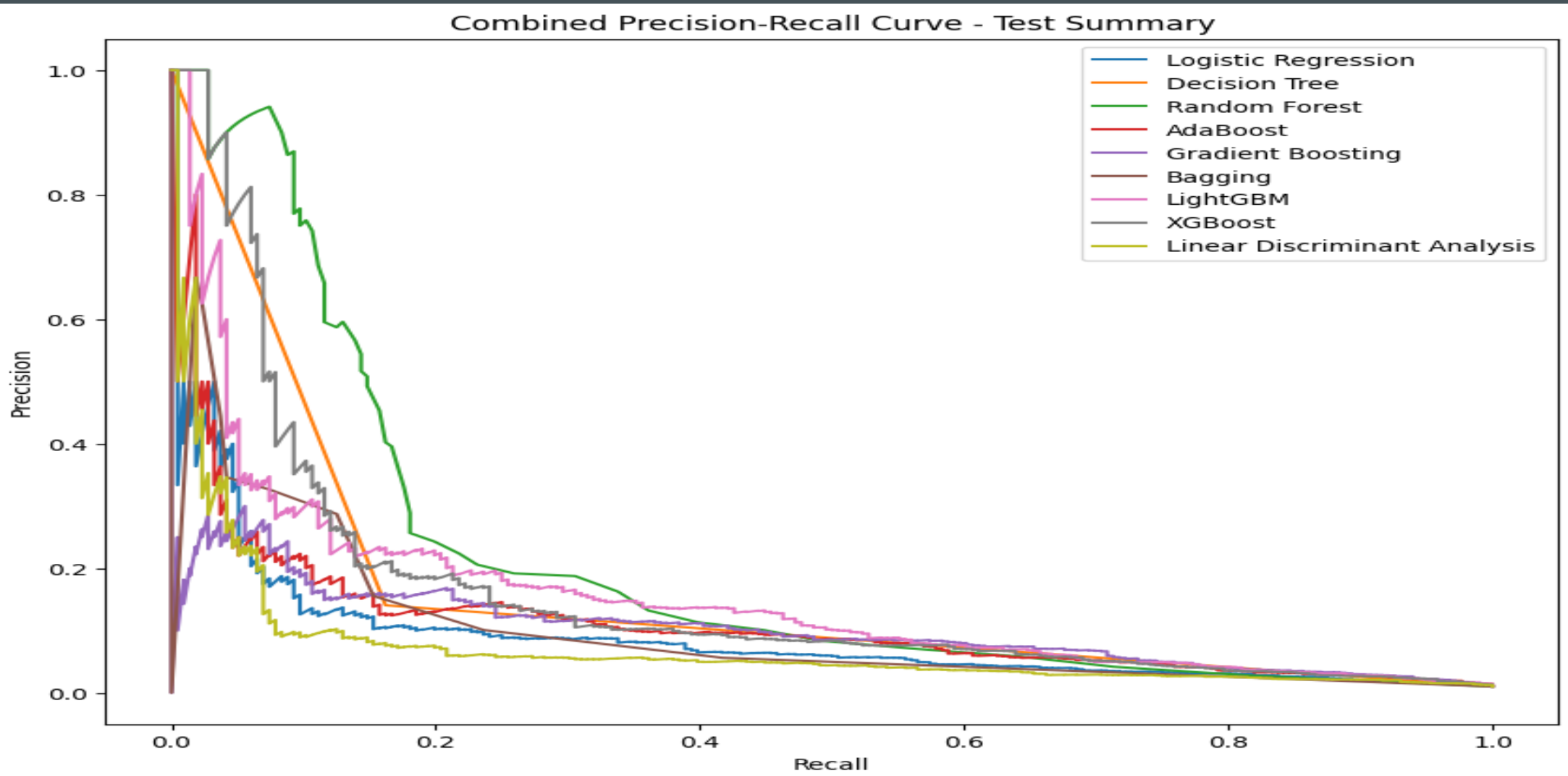

Figure 10: Combined Precision - recall curve for test summary – All models

TABULAR SUMMARY OF ALL MODELS

S.No	Name of the model	Test Accuracy
1	Random Forest Classifier model	99.0435
2	XGB Classifier model	99.006
3	Logistic Regression model	98.9919
4	AdaBoost Classifier model	98.9544
5	Bagging Classifier model	98.9497
6	Gradient Boosting Classifier model	98.9075
7	Light GBM Classifier model	98.9075
8	Linear Discriminant Analysis model	98.1949
9	Decision Tree Classifier model	98.148

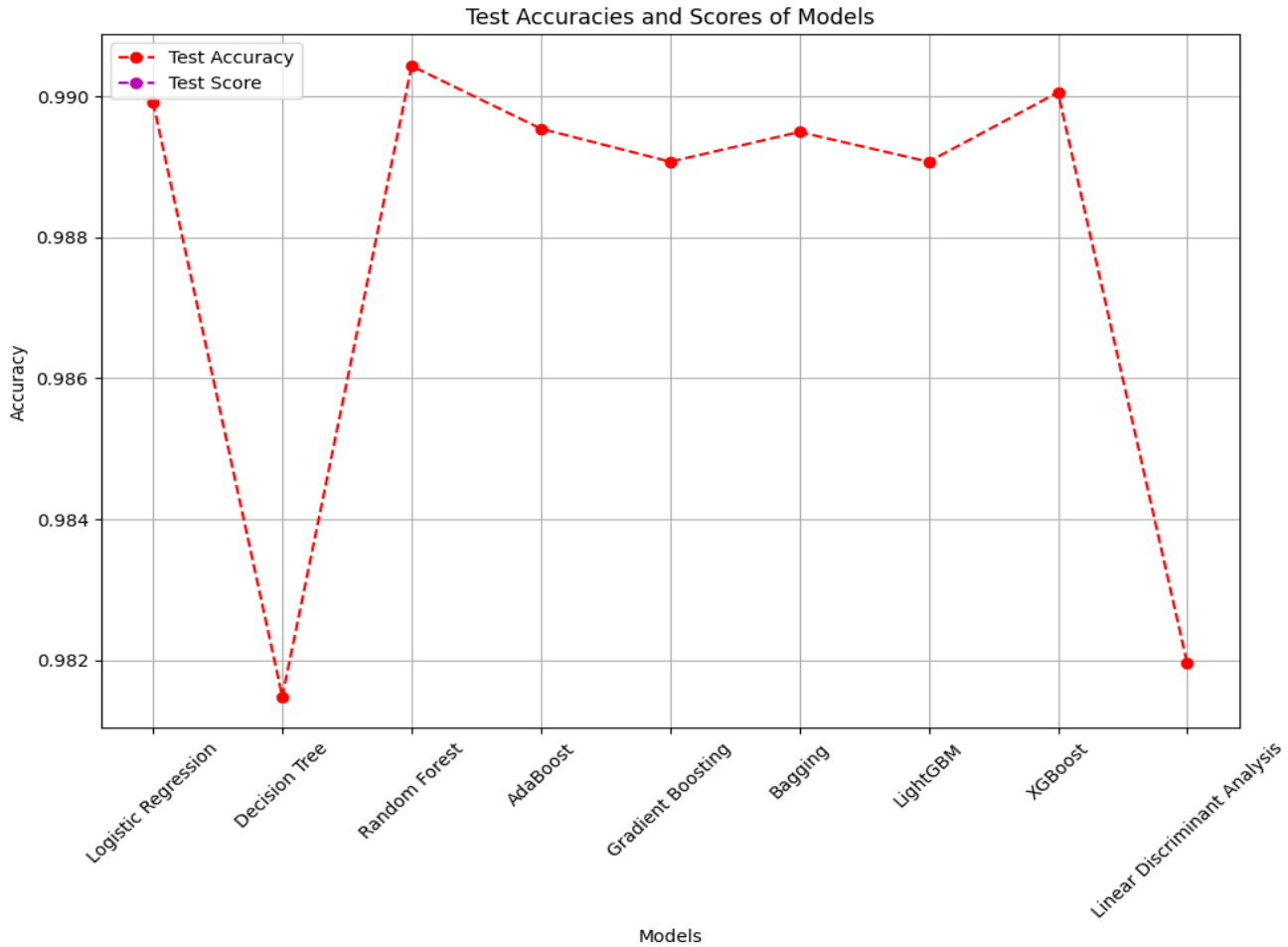

Figure 11: Test accuracy comparison for all models

MODEL TUNING FOR BEST MODEL

- Post testing all the models and from the results, Random forest classifier is the best model with test accuracy - 0.9904355572225
- Applying GridsearchCV for model tuning to random forest model
- From the results it has been observed the post model tuning the test accuracy is decreased to 0.9902949036523043
- In this case, the difference in accuracy is minimal, indicates that the initial Random Forest model was performing very well and not required significant tuning.

```
Random Forest Model Tuning Results:
Best Hyperparameters: {'max_depth': None, 'n_estimators': 50}
Best Model Accuracy: 0.9902949036523043
Classification Report:
              precision    recall  f1-score   support

         0.0       0.99      1.00      1.00     21113
         1.0       0.85      0.05      0.10       216

    accuracy                           0.99     21329
   macro avg       0.92      0.53      0.55     21329
weighted avg       0.99      0.99      0.99     21329
```

INSIGHTS FROM THE MODELS

- **Logistic Regression:** High accuracy, but low recall for defaulters suggests limited ability to identify default instances.
- **Decision Tree:** Accurate in identifying non-defaulters, but limited ability to identify defaulters due to low precision and recall.
- **Random Forest:** High accuracy with decent precision for non-defaulters, but limited ability to identify defaulters.
- **AdaBoost:** Accurate in identifying non-defaulters, but limited precision and recall for defaulters.
- **Gradient Boosting:** Good accuracy for non-defaulters, but limited ability to identify defaulters due to low precision and recall.
- **Bagging:** Accurate in identifying non-defaulters, moderate precision for defaulters, but limited recall.
- **Light GBM:** Accurate in identifying non-defaulters, limited precision and recall for defaulters.
- **XG Boost:** Accurate in identifying non-defaulters, some ability to identify defaulters but with limitations.
- **Linear Discriminant Analysis:** Accurate in identifying non-defaulters, but limited precision and recall for defaulters.

BUSINESS RECOMMENDATIONS

- **Utilize Predictive Models:** Enhance risk management using predictive models for credit card operations.
- **Improve Credit Assessment:** Leverage model insights for better decision-making in credit assessment.
- **Efficient Resource Allocation:** Allocate resources wisely based on risk insights to maximize profitability.
- **Proactive Customer Protection:** Implement proactive measures to safeguard customers from default risks.
- **Enhance Competitiveness:** Present proactive risk management strategies to stay competitive and retain customers.

Thank you